

Software Development Life Cycle (SDLC)

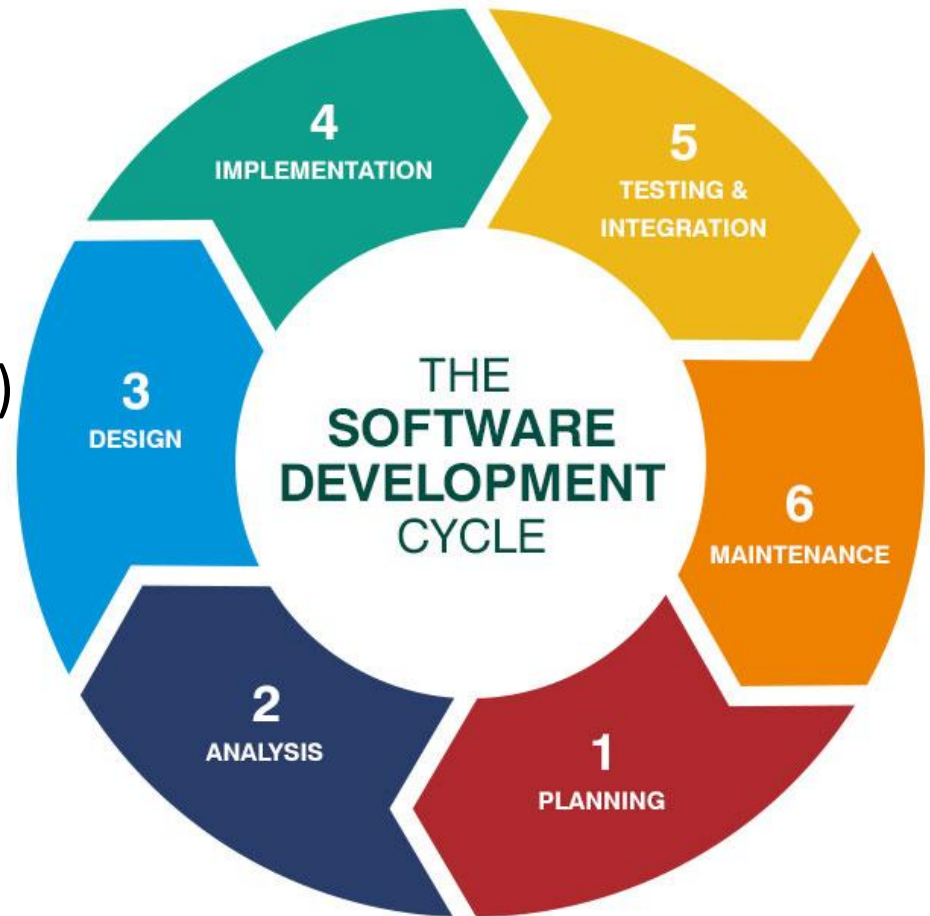
- A systematic process used to develop high-quality software.
- It defines phases involved in planning, creating, testing, and maintaining software.

What is SDLC?

- SDLC is a framework that describes all phases involved in software development.
- It ensures structured and organized development.

SDLC Phase's

1. Requirement Analysis
2. System Design
3. Implementation (Coding)
4. Testing
5. Deployment
6. Maintenance



Phase 1: Requirement Analysis

- Collect requirements from users or clients.
- Understand what the system should do.
- Document all requirements clearly.

Phase 2: System Design

- Design system architecture.
- Prepare database design.
- Plan UI and system structure.

Phase 3: Implementation (Coding)

- Developers write actual code.
- Use programming languages to build software.
- Convert design into working system.

Phase 4: Testing

- Test software for bugs and errors.
- Perform different testing types.
- Ensure software works correctly.

Phase 5: Deployment

- Release software to users.
- Install system in real environment.
- Make it available for actual use.

Phase 6: Maintenance

- Fix bugs after deployment.
- Improve features.
- Update system when needed.

Conclusion

- SDLC helps in structured development.
- Improves software quality.
- Ensures proper planning, development, testing, and maintenance.